

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Industry Problem Solving	Assessment:	Assessment Tool 2– Industry Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	A. Reddy Prasad	Reg. No.:	192472045
Section / Batch:	2024	Date:	20-08-2026

Code of Conduct

I A. Reddy Prasad (192472045) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A. Reddy Prasad

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

		analyze the case deeply				
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

6. Software Industry — API Response Time

Given

- Sample size,
- Sample mean, ms
- Sample standard deviation, ms
- Confidence level = 95%
- Since , use the normal approximation with .

1. Population and Sample

- **Population:** Response times of all API requests handled by the software company's system.
- **Sample:** 100 randomly selected API requests.

2. Parameter of Interest

Population mean API response time, .

3. Point Estimate

4. Null Hypothesis

5. Alternative Hypothesis

6. Significance Level

7. 95% Confidence Interval

Formula:

Therefore,

8. Test Statistic and p-value

Two-tailed p-value:

9. Compare p-value with α

Therefore, **fail to reject**.

10. Industry-Oriented Conclusion

The software company can be 95% confident that the true average API response time is between 233.14 ms and 246.86 ms. The estimated average response time is approximately 240 ms, indicating that the API generally operates around the 240 ms level.

7. E-Commerce — Conversion Rate

Given

- Sample size
- Number of purchasers
- Confidence level = 95%

1. Population and Sample

- **Population:** All visitors to the e-commerce platform.
- **Sample:** 600 randomly selected visitors.

2. Parameter of Interest

Population conversion rate, .

3. Point Estimate

Therefore,

4. Null Hypothesis

For the required hypothesis format, using the estimated rate as the reference:

5. Alternative Hypothesis

6. Significance Level

7. 95% Confidence Interval

Formula:

Therefore,

or

8. Test Statistic and p-value

9. Compare p-value with α

Therefore, **fail to reject**.

10. Industry-Oriented Conclusion

The e-commerce company can be **95% confident that the true conversion rate lies between 66.33% and 73.67%**. The estimated conversion rate is **70%**, indicating a relatively strong purchasing rate among visitors.

8. Telecom — Monthly Data Consumption

Given

- Sample mean GB
- Sample standard deviation GB
- Confidence level = 99%

1. Population and Sample

- **Population:** Monthly data consumption of all users of the telecom company.
- **Sample:** 225 randomly selected users.

2. Parameter of Interest

Population mean monthly data consumption, .

3. Point Estimate

4. Null Hypothesis

5. Alternative Hypothesis

6. Significance Level

For a 99% confidence level:

7. 99% Confidence Interval

Therefore,

Approximately:

8. Test Statistic and p-value

9. Compare p-value with α

Therefore, **fail to reject** .

10. Industry-Oriented Conclusion

The telecom company can be 99% confident that the true average monthly data consumption is between 14.03 GB and 15.57 GB. This information can help the company plan data capacity, network resources, and suitable data plans.

9. Manufacturing — Component Diameter

Given

- Historical population mean = 50 mm
- Sample mean mm
- Sample standard deviation mm
- Confidence level = 95%

Because population standard deviation is unknown, use a **t-distribution**.

Degrees of freedom:

For 95% CI:

1. Population and Sample

- **Population:** All components produced by the machine.
- **Sample:** 36 randomly selected components.

2. Parameter of Interest

Population mean component diameter, .

3. Point Estimate

4. Null Hypothesis

5. Alternative Hypothesis

This is a **two-tailed test** because we want to know whether the mean differs from 50 mm in either direction.

6. Significance Level

7. 95% Confidence Interval

Formula:

Therefore,

8. Test Statistic and p-value

For , the two-tailed p-value is approximately:

9. Compare p-value with α

Therefore, **fail to reject** .

Also notice:

lies inside the 95% confidence interval:

10. Industry-Oriented Conclusion

The value 50 mm is a plausible population mean because it lies within the 95% confidence interval. At the 5% significance level, there is not enough statistical evidence to conclude that the machine's true average diameter differs from 50 mm.

10. Banking — Loan Processing Time

Given

- Claimed mean = 5 days
- Sample mean days
- Sample standard deviation days
- Confidence level = 95%

Since population standard deviation is unknown, use the **t-distribution**.

1. Population and Sample

- **Population:** Processing times of all loan applications handled by the bank.

- **Sample:** 49 randomly selected loan applications.

2. Parameter of Interest

Population mean loan-processing time, μ .

3. Point Estimate

4. Null Hypothesis

The bank claims the average is 5 days:

5. Alternative Hypothesis

6. Significance Level

7. 95% Confidence Interval

Therefore,

Approximately:

8. Test Statistic and p-value

For μ , two-tailed:

9. Compare p-value with α

Therefore, **reject** H_0 .

Also, 5 days is **outside the 95% confidence interval**:

10. Industry-Oriented Conclusion

There is statistically significant evidence that the bank's true average loan-processing time is different from 5 days. Since the sample mean is 5.6 days and the entire confidence interval is above 5 days, the evidence suggests that the bank's actual average processing time is longer than its claimed 5 days.